

MAT1002 Lecture 17,

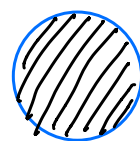
Outline

- Optimization with equality constraints
 - ↳ Lagrange multiplier
- Results related to second partials

Optimization with Equality Constraints

Q: Given a mountain altitude function ^{cts} $f(x,y)$, what is the highest altitude given by points in the unit disk $x^2 + y^2 \leq 1$?

- Last time: find crit. pts. in the interior of D , and find the maximum in the boundary, then compare.



$$D: x^2 + y^2 \leq 1$$

Notation

$$\begin{array}{ll} \text{Maximize} & f(x,y) \\ \text{Subject to} & \underline{x^2 + y^2 = 1.} \end{array} \quad \leftarrow \text{Boundary}$$

(s.t.)

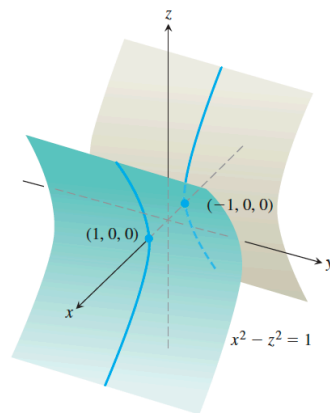
This means we are trying to find $\max_{(x,y) \in C} f(x,y)$, where

$$C = \{(x,y) : x^2 + y^2 = 1\}.$$

Another example Find all points on the surface $x^2 - z^2 - 1 = 0$ that are closest to the origin.

Notation

$$\begin{array}{ll} \text{Minimize} & \sqrt{x^2 + y^2 + z^2} \\ \text{Subject to} & x^2 - z^2 - 1 = 0. \end{array}$$



Optimization with One Equality Constraint

$$\begin{array}{ll} \text{Maximize (or minimize)} & f(x_1, \dots, x_n) \\ \text{Subject to} & g(x_1, \dots, x_n) = 0 \\ & \text{(s.t.)} \end{array}$$

Consider Minimize $x^2 + y^2 + z^2$ s.t. $x^2 - z^2 - 1 = 0$. S

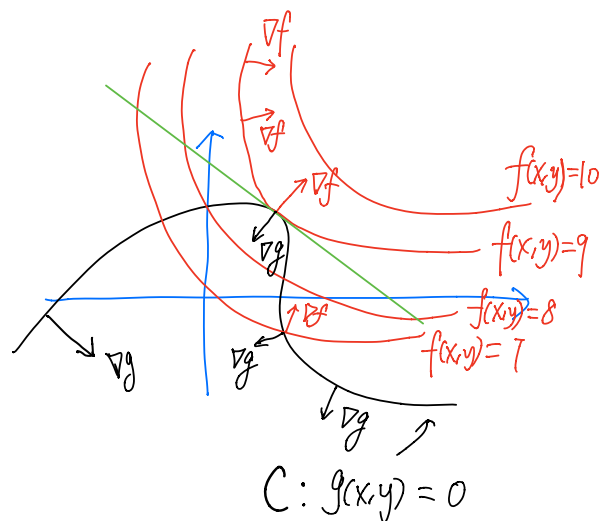
Critical pt method will not help, since for $f(x, y, z) = x^2 + y^2 + z^2$,
 $f_x = 2x$, $f_y = 2y$, $f_z = 2z$, so the only critical pt is
 $(x, y, z) = (0, 0, 0)$, which is not in S !

Need another method.

Method of Lagrange Multipliers

Consider a 2-variable case:

$$\begin{array}{ll} \text{Max} & f(x, y) \\ \text{s.t.} & g(x, y) = 0 \end{array}$$



Theorem (14.8.12, 2D Version)

Let $f(x,y)$ be a differentiable function on D , and let C given by

$$\vec{r}(t) = \langle x(t), y(t) \rangle$$

be a ^{Smooth} curve contained in the interior of D . If P_0 is an interior point of C that gives a local extremum of f relative to points in C , then $\nabla f \perp \vec{r}'$ at P_0 .

This thm is also true for $f(x,y,z)$ and $C: \vec{r}(t) = \langle x(t), y(t), z(t) \rangle$

Proof: • Consider the values of f , but only on C .

• Let $g(t) := f(\underbrace{x(t), y(t)}_{\in C})$, and let $P_0 = (x(t_0), y(t_0))$.

• By assumption, $f(x(t_0), y(t_0))$ is a local extremum of f on C , so $g(t_0)$ is extremum on $(t_0 - \delta, t_0 + \delta)$ for some $\delta > 0$. Hence $g'(t_0) = 0$.

• But $g'(t_0) = f_x(P_0)x'(t_0) + f_y(P_0)y'(t_0) = \nabla f(P_0) \cdot \vec{r}'(t_0)$,

So

$$\nabla f(P_0) \cdot \vec{r}'(t_0) = 0.$$

interior of C □

In particular, when f has a local extremum at a point P_0 relative to C , where C is $g(x,y) = 0$, then both ∇f and g is differentiable

Recall the gradient vector, ∇g , is orthogonal to its level surface.

∇g are $\perp$ to the curve C at P_0 , so either:

- $\nabla g = \langle 0, 0 \rangle$, or ;
- $\nabla f = \lambda \nabla g$ for some $\lambda \in \mathbb{R}$ ($\nabla f \parallel \nabla g$).

This observation suggests the following method.

The Method of Lagrange Multipliers

Suppose that the minimum of $f(x, y)$ subject to the constraint $g(x, y) = 0$ exists, where f and g are differentiable. Assume that $\nabla g(x, y) \neq \vec{0}$ whenever $g(x, y) = 0$. The following is a procedure for finding the constrained minimum:

1. Find all values of x , y and λ that simultaneously satisfy

$$\nabla f(x, y) = \lambda \nabla g(x, y) \quad \text{and} \quad g(x, y) = 0.$$

2. Evaluate f at all the points (x, y) obtained in step 1. The smallest value is the desired minimum.

If the constrained maximum exists, then the largest value in step 2 is the maximum.

Example

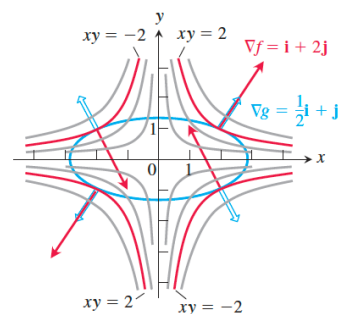
Find the maximum and minimum of the function

$$f(x, y) := xy$$

defined on the ellipse

$$\frac{x^2}{8} + \frac{y^2}{2} = 1.$$

Specify where the extrema occur on the ellipse.



Ans: Max at $(2, 1)$ and $(-2, -1)$; Min at $(2, -1)$ and $(-2, 1)$.

The method also works for functions with more variables.

e.g. Find all points on the surface $S: x^2 - z^2 - 1 = 0$ that are closest to origin, and find the minimum distance.

Ans: $(\pm 1, 0, 0)$; distance = 1.

Two Equality Constraints

Consider Max/min $f(x, y, z)$
s.t. $g_1(x, y, z) = 0$ and $g_2(x, y, z) = 0$,

that is, finding extrema of f on the curve $S_1 \cap S_2$, where

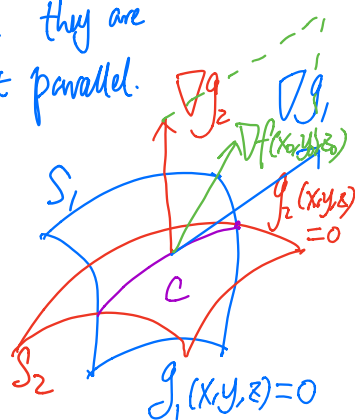
$$S_i = \{(x, y, z) : g_i(x, y, z) = 0\}, \quad i \in \{1, 2\}.$$

Surface

If f has a local extremum at $P_0 \in S_1 \cap S_2$ relative to points in $S_1 \cap S_2$, then $\nabla f(P_0)$ must be $\perp$ to $C = S_1 \cap S_2$. Since ∇g_1 and ∇g_2 are also $\perp$ to C , we have ∇f lying in the plane spanned by $\nabla g_1(P_0)$ and $\nabla g_2(P_0)$. Assuming they are $\neq \vec{0}$, not parallel.

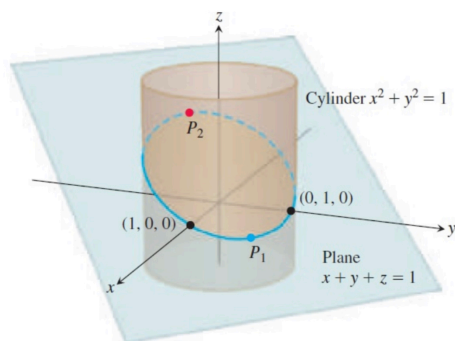
This means that the method of Lagrange multipliers can still be used, but the system to be solved is the following instead:

$$\begin{cases} \nabla f(x, y, z) = \lambda_1 \nabla g_1(x, y, z) + \lambda_2 \nabla g_2(x, y, z) \\ g_1(x, y, z) = 0 \\ g_2(x, y, z) = 0 \end{cases}$$



Example 14.8 Example 5

The plane $x + y + z = 1$ cuts the cylinder $x^2 + y^2 = 1$ in an ellipse. Find the points on the ellipse that lie closest to and farthest from the origin.



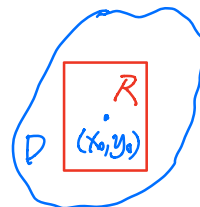
Ans: Farthest: $(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 1-\sqrt{2})$ and $(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 1+\sqrt{2})$.
Closest: $(0, 1, 0)$ and $(1, 0, 0)$.

Proof of Some Results Regarding Second-Order Partial

1. Error of Standard Linear Approximation

Result in short: $|E(x, y)| \leq \frac{1}{2}M(|x-x_0| + |y-y_0|)^2$, where M bounds the absolute values of all second partials on R .

(All second partials are cts.) ($E(x, y) = f(x, y) - L(x, y)$)



Proof: • Let $h := \Delta x$ and $k := \Delta y$ be "small changes".

• Define $F(t) := f(x_0 + th, y_0 + tk)$, $t \in [0, 1]$.

• By Taylor's theorem, since f is cts on $[0, 1]$ & diff'able on $(0, 1)$,

$$F(1) = F(0) + F'(0)(1-0) + \frac{1}{2}F''(c)(1-0)^2$$

$$= F(0) + F'(0) + \frac{1}{2}F''(c), \text{ where } 0 < c < 1 \quad \textcircled{1}$$

• $F'(t) = f_x'x'(t) + f_y'y'(t)$ (Chain rule, F diff'able.)

$$= f_x h + f_y k.$$

• Since all second partials of f are cts, F' is diff'able.

• $F''(t) = \frac{dF'}{dt} = F'_x x'(t) + F'_y y'(t)$ (Chain rule)

$$= (f_x h + f_y k)_x h + (f_x h + f_y k)_y k$$

$$= f_{xx} h^2 + 2f_{xy} h k + f_{yy} k^2$$

• Now $\textcircled{1} \Rightarrow$ $L(x_0+h, y_0+k)$

$$F(1) = f(x_0+h, y_0+k) = \underbrace{f(x_0, y_0) + f_x(x_0, y_0)h + f_y(x_0, y_0)k}_{E(x_0+h, y_0+k)} + \underbrace{\frac{1}{2}(f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2)}_{E(x_0+h, y_0+k)} \Big|_{(x,y)=(x_0+h, y_0+k)}$$

• In short notation, and by triangle inequality,

$$|E(x,y)| = \frac{1}{2} |f_{xx}(x-x_0)^2 + 2f_{xy}(x-x_0)(y-y_0) + f_{yy}(y-y_0)^2|$$

$$\leq \frac{1}{2} \left(\underbrace{|f_{xx}|}_{\downarrow} (x-x_0)^2 + 2 \underbrace{|f_{xy}|}_{\swarrow} |x-x_0| |y-y_0| + \underbrace{|f_{yy}|}_{\nwarrow} (y-y_0)^2 \right)$$

$$\leq \frac{1}{2} M (|x-x_0| + |y-y_0|)^2.$$

